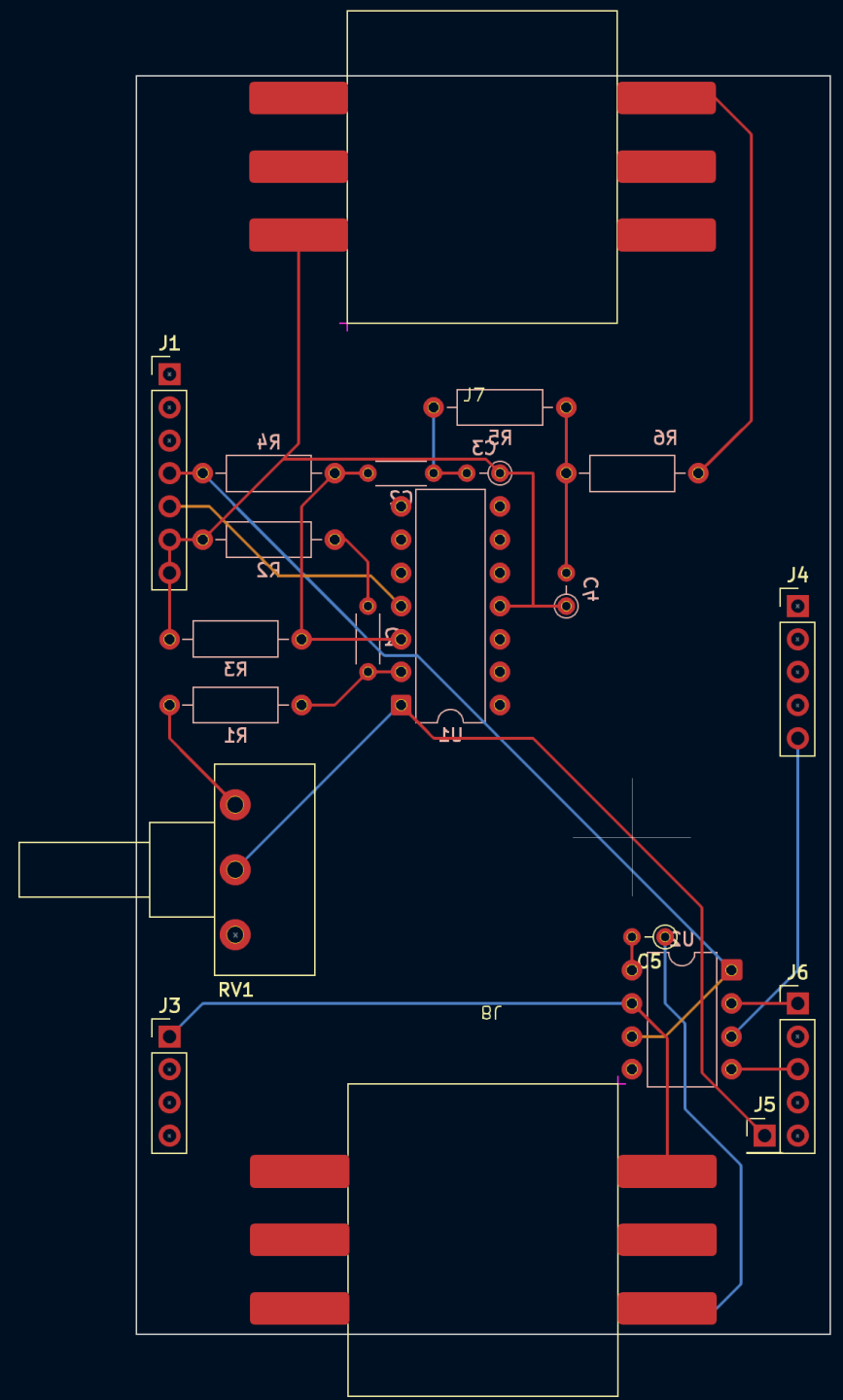


REPEAT-INATOR: DIGITAL GUITAR DELAY PEDAL

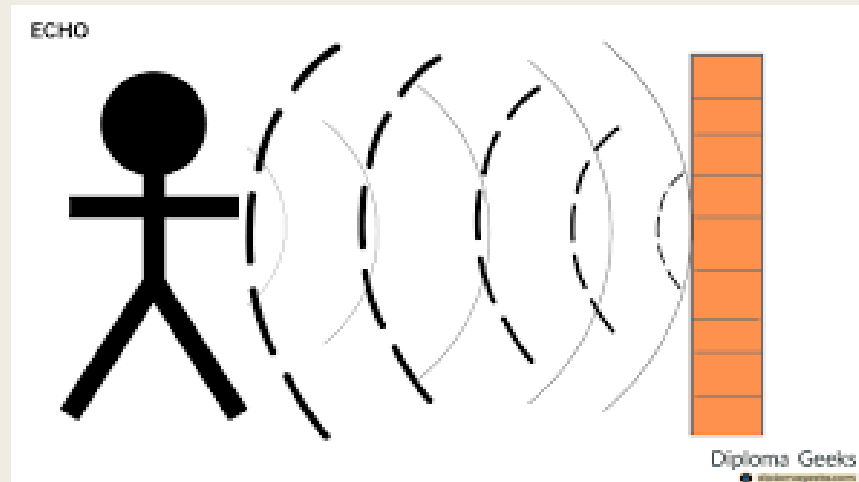
EE-329 – Custom Project

Cameron Young, Maddie Masiello,
Greg Karaoghlanian

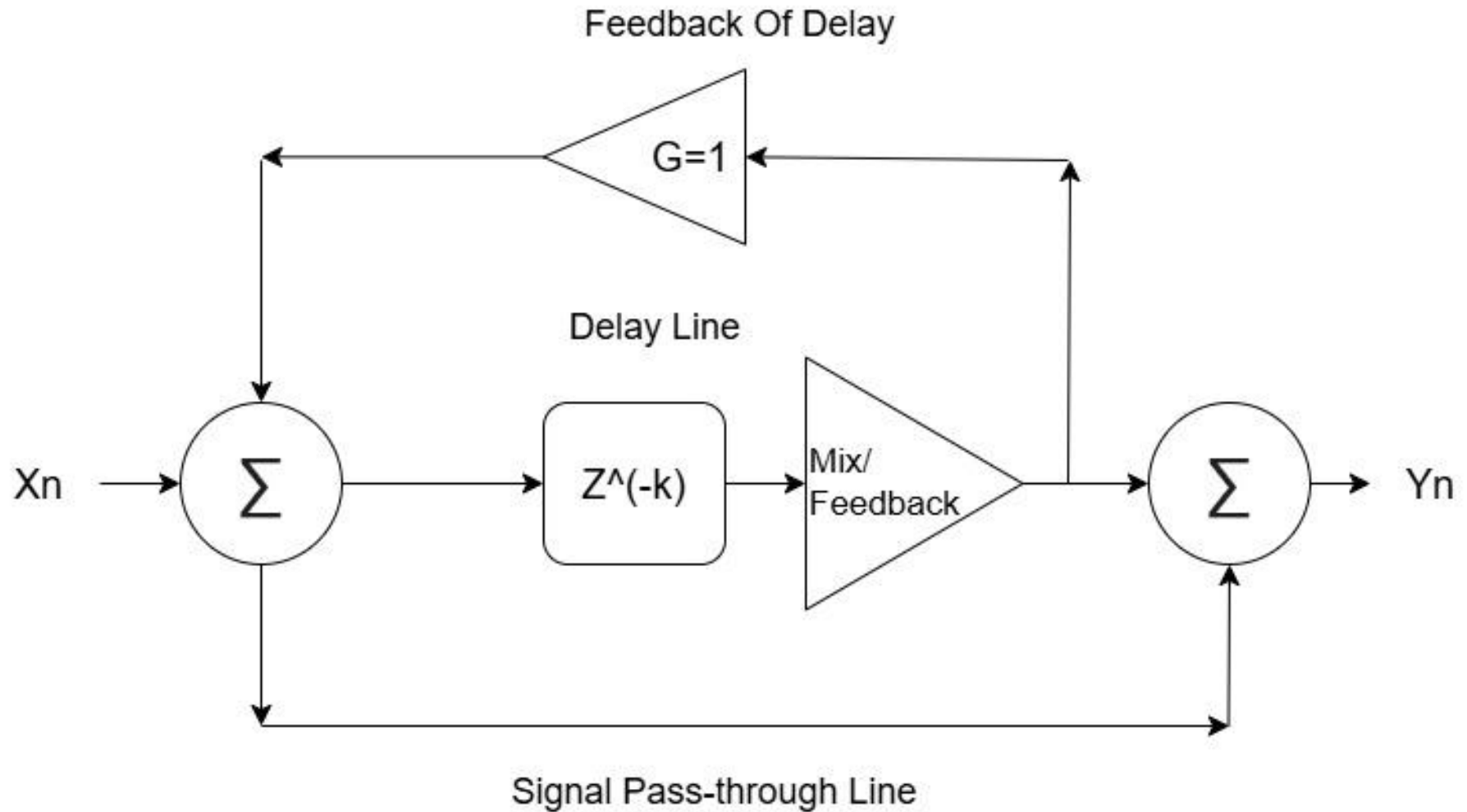


What is the Repeat-inator?

- Custom built digital guitar delay effect pedal
- Processes analog input signal and creates adjustable echo effects
- Adjustable delay time, mix/feedback, output volume



System Design Highlights



System Design Highlights – Circular Buffer

- DSP operation applies delay effect to an audio signal
- Takes current input sample (DSP_sample)
- Delayed sample scaled by an adjustable factor of 0.5
- Adds delayed version of a past sample from buffer & stores

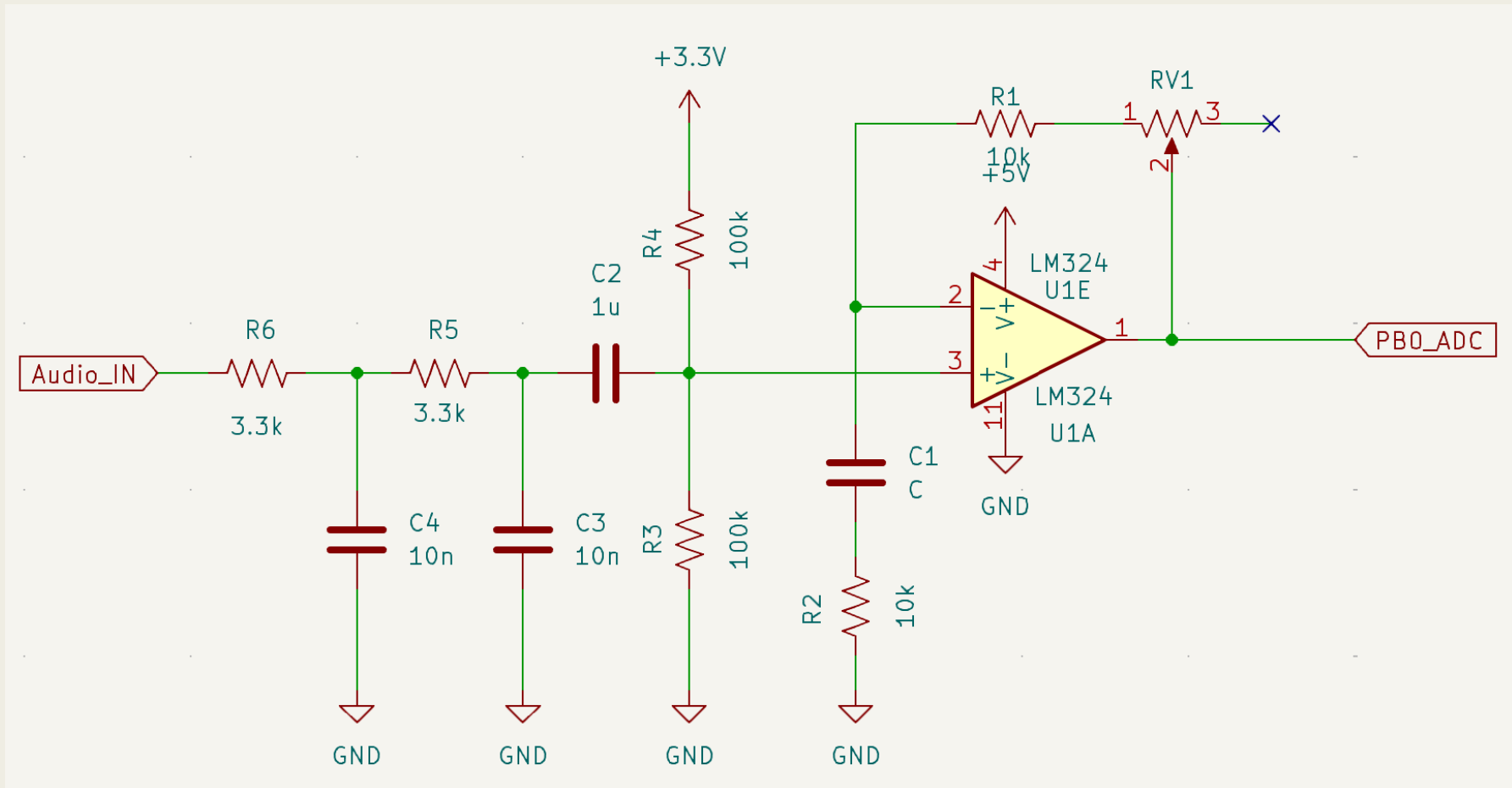
```
output_sample = current_sample + (delay_line[read_index] * 0.5)
delay_line[write_index] = output_sample;
write_index = (write_index + 1) % buffer_size
```

System Design Highlights – UART

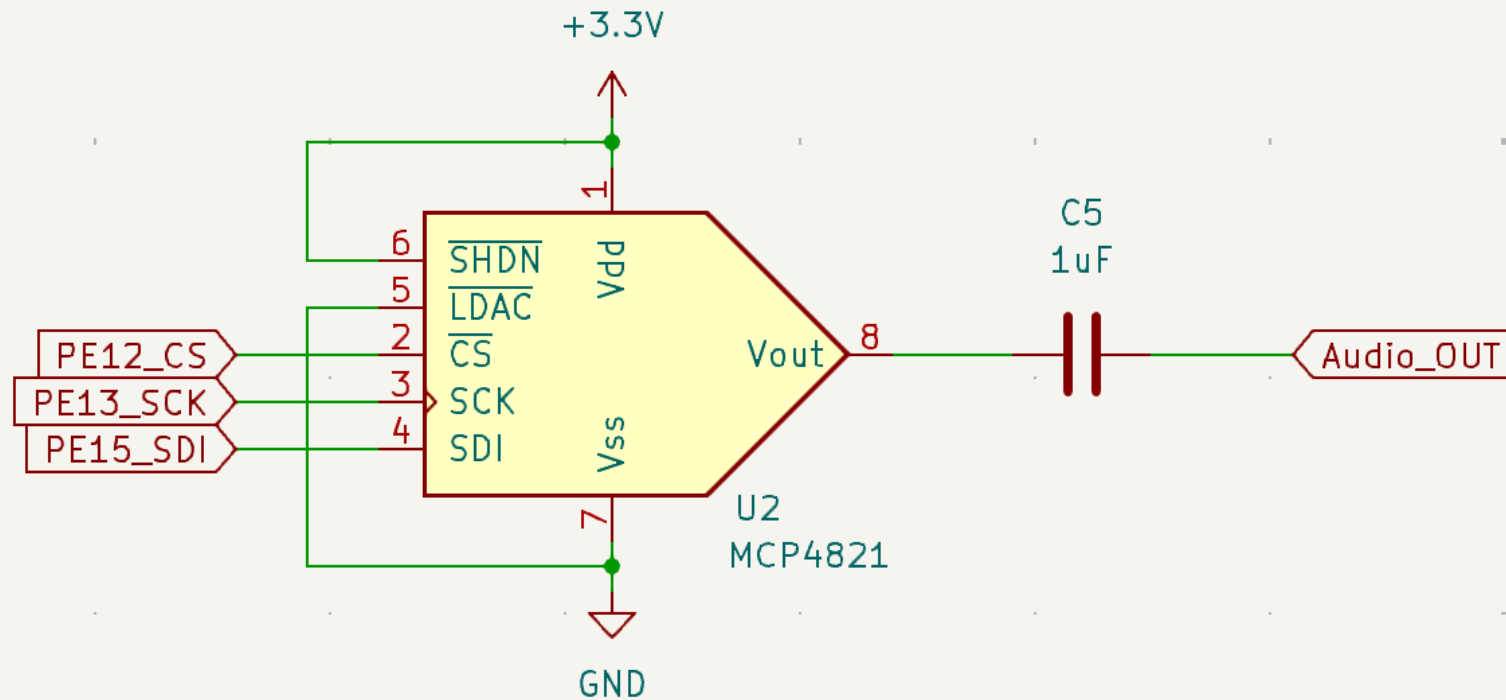
- Uses LPUART1 on PG7 (TX) and PG8 (RX)
- 115200 baud rate
- volume, mix, & delay real-time modifiable variables
- "G", "R", "V", "M", & "D"

```
Current settings:
Volume: 100%
Mix: 50%
Delay: 42000 samples
Parameters reset to defaults
Volume now set to: 100%
Mix now set to: 50%
Delay now set to: 42000 samples
Enter volume (0-100): 60
Volume now set to: 60%
Enter mix (0-100): 20
Mix now set to: 20%
Enter delay (0-50000 samples): 45000
Delay now set to: 45000
Current settings:
Volume: 60%
Mix: 20%
Delay: 45000 samples
```

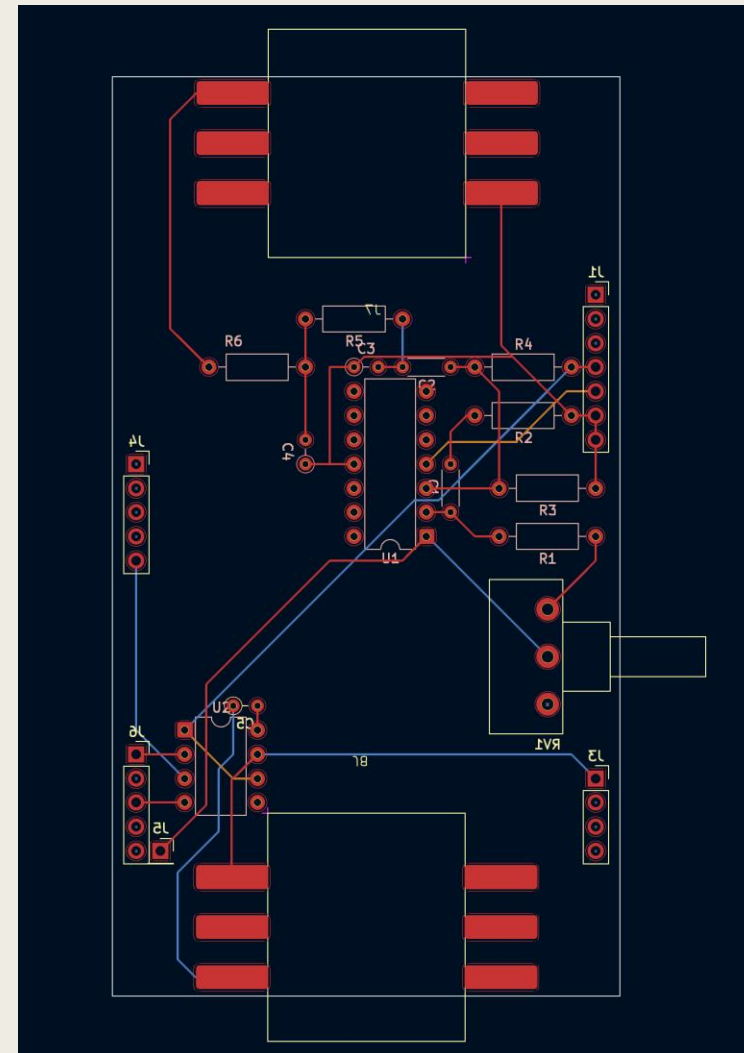
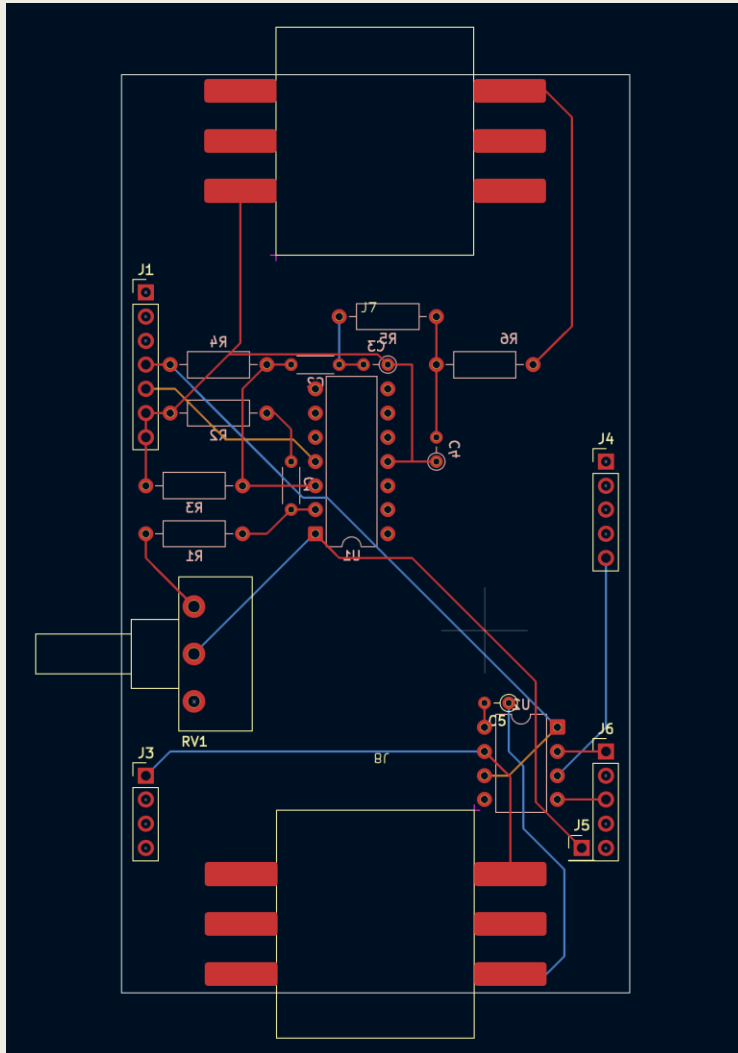
System Design Highlight – Input Stage: Filter Op-amp



System Design Highlight – Output Stage: DAC

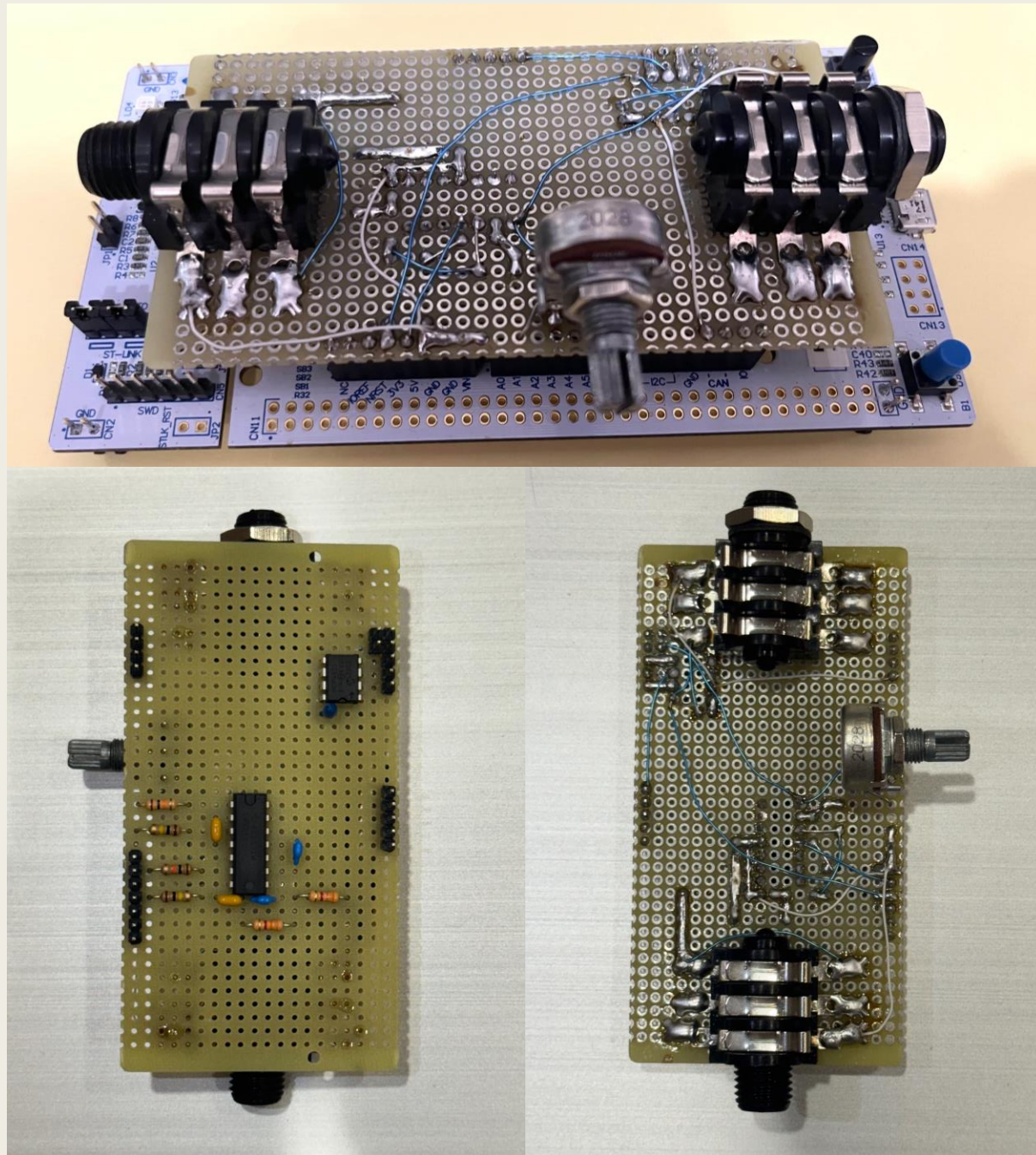


Perf-board Layout



Hardware components

- Potentiometer
- LM324 Opamp
- ¼ inch Audio Jacks
- DAC
- UART communication (peripheral)



Technical Specifications

- Sample Rate: 40 kHz, 12-bit ADC and DAC
- Max Delay: 1 second (around 50,000 samples)
- Power Consumption: <100 mW
- Output Range: 0–2.048V; Input Range: 0–3.3V
- Latency: <1 ms processing time



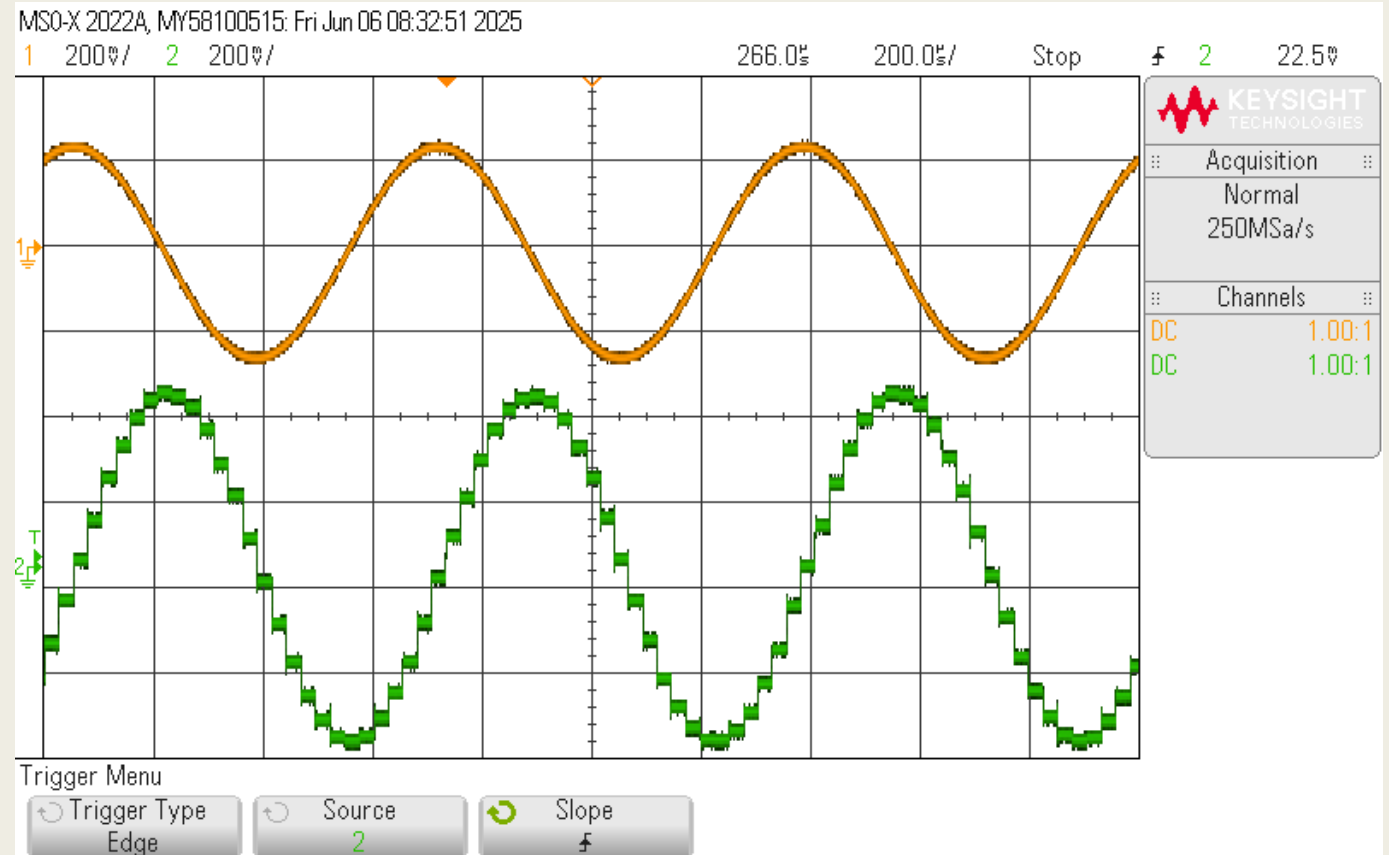
Demo

- Signal passthrough verification
- Delay algorithm functional w/ live guitar input
- UART
- DAC outputs scaled echo signal to amp



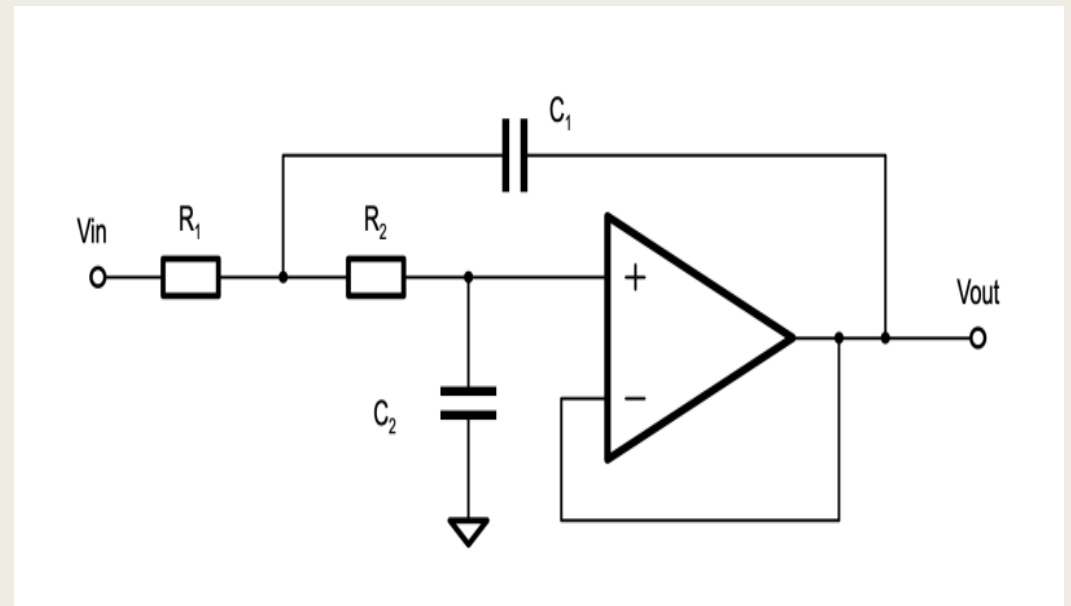
Obstacles & Lessons Learned

- Noise & aliasing
- Opamp + DAC wiring w/ Nucleo board
- Perf board hardware issues
- Amplitude overflow
- DC bias scaling



Future Functionality

- Adjustable sample rate
- On/off button (hardware)
- RC low-pass filter on output
- Sallen-Key active low pass filter input (combines filter and op-amp)
- UART



References

- [1] LEGO Group, “Strike a chord with the new LEGO® Ideas Fender® Stratocaster® set,” LEGO Newsroom, Sep. 13, 2021. [Online]. Available: <https://www.lego.com/en-us/aboutus/news/2021/september/lego-ideas-fender-stratocaster-set>
- [2] Diploma Geeks, “Define echo and its applications,” *Diploma Geeks*, Feb. 2023. [Online]. Available: <https://www.diplomageeks.com/2023/02/define-echo-and-its-applications.html>
- [3] R. Lacoste, "Sallen-&-Key Filter," *Circuit Cellar*, Feb. 27, 2023. [Online]. Available: <https://circuitcellar.com/resources/quickbits/sallen-n-key-filter/>
- [4] Sweetwater Sound, “Boss DD-8 Digital Delay Pedal,” *Sweetwater*, [Online]. Available: <https://www.sweetwater.com/store/detail/DD8-boss-dd-8-digital-delay-pedal>. [Accessed: Jun. 5, 2025].
- [5] National Academy of Engineering, "Harry Nyquist," *National Academy of Engineering*, 2021. [Online]. Available: <https://www.nae.edu/238067/Harry-Nyquist>